taking place.

It is certainly no wonder that the discovery of this law—the so-called least-action principle, after which the elementary quantum of action was later also named,—made its discoverer Leibniz, and soon after him also his follower Maupertuis, so boundlessly enthusiastic; for these scientists believed themselves to have found in it a tangible evidence for a ubiquitous higher reason ruling all nature.

In fact, the least-action principle introduces a completely new idea into the concept of causality: The *causa efficiens*, which operates from the present into the future and makes future situations appear as determined by earlier ones, is joined by the *causa finalis* for which, inversely, the future—namely, a definite goal—serves as the premise from which there can be deduced the development of the processes which lead to this goal.

So long as we confine ourselves to the realm of physics, these alternative points of view are merely different mathematical expressions for one and the same fact, and it would be futile to ask which of the two came nearer to the truth. The choice between them depends solely on practical considerations. The chief advantage of the least-action principle is that it requires no definite frame of reference for its formulation. This principle is therefore excellently adapted for carrying through transformation of coordinates.

But we are now interested in questions of a more general character. It will suffice at this point to note only that the historical development of theoretic research in physics has led in a remarkable way to a formulation of the principle of physical causality which possesses an explicitly teleological character; but at the same time this formulation introduces nothing substantially new or even contradictory into the character of the laws of nature. The issue is simply one of different perspectives of interpretation, both of which are equally well justified by the actual facts. The situation in biology should be no different than we have found it in physics, although in biology the difference between the two viewpoints has assumed far sharper outlines.

In any case, we may say in summary that according to what exact natural science teaches us, the entire realm of nature, in which we human beings on our tiny mote of a planet play only an infinitesimally small part, is ruled by definite laws which are independent of the existence of thinking human beings; but these laws, insofar as they can at all be comprehended by our senses, can be given a formulation which is adapted for purposeful activity. Thus, natural science exhibits a rational world order to which nature and mankind are subject, but a world order the inner essence of which is and remains unknowable to us, since only our sense data (which can never be completely excluded) supply evidence for it. Nevertheless, the truly prolific results of natural-scientific research justify the conclusion that continuing efforts will at least keep bringing us progressively nearer to the inattainable